

An integrated automated inspection system using computer vision and real-time monitoring

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ABSTRACT

Traditional quality inspection on industrial production lines relies heavily on dedicated sensor arrays, mechanical switches, and rule-based vision systems. These approaches are costly to deploy, brittle under varying conditions, and incapable of generalising to new product types without significant re-engineering. This paper presents a computer vision system designed to replace or supplement such sensor-heavy pipelines using YOLO, a state-of-the-art real-time object detection framework. Using paper boxes as a representative case study for logistics and packaging inspection, the system classifies products as valid or damaged directly from a camera feed mounted over a conveyor belt with detected defective items ejected from the production line. The system is supervised through a web-based dashboard providing live feed monitoring, system control, and automated PDF production reports. This paper describes the full pipeline from data collection and augmentation strategy to model training, and software architecture. A controlled experiment comparing augmented versus non-augmented training sets, demonstrate the critical role of augmentation in building robust, generalisable inspection models. Results and implications for future deployment in SCADA-like industrial control environments are discussed.

Keywords: computer vision, data augmentation, industrial AI, SCADA, YOLO.

1. INTRODUCTION

Modern manufacturing and logistics facilities depend on automated inspection to maintain product quality and minimise human error. Conventional approaches rely on photoelectric sensors, mechanical jigs, or rule-based image-processing systems built on handcrafted features such as edge detection and template matching. While effective in highly controlled settings, these systems require separate hardware per inspection attribute, are expensive to reconfigure for new products, and degrade readily when lighting or product appearance varies.

Computer vision offers a fundamentally different model: a single camera, paired with a trained neural network, can simultaneously detect product presence, classify type, assess structural integrity, and trigger downstream actuation. This multi-attribute inspection from one sensor is the central advantage of vision-based automation over traditional sensor arrays, and its applicability spans industries from food and beverage, automotive, and electronics manufacturing to recycling, pharmaceuticals, and agriculture [1].

Among available detection frameworks, the You Only Look Once (YOLO) family has become the dominant choice for real-time industrial inspection due to its favourable trade-off between speed and accuracy [2]. YOLO11, released by Ultralytics in 2024, introduces an anchor free architecture and a decoupled detection head that improves both accuracy and training

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convergence relative to earlier versions [3]. Prior work has demonstrated YOLO suitability across surface defect detection [4], printed circuit board (PCB) inspection [5], and food grading [6].

A persistent challenge in industrial vision is data scarcity: capturing sufficient labelled images of rare defect types is time-consuming and costly. Augmentation strategies have been shown to improve generalisation substantially [7] and are particularly critical when the dataset contains color patterns that could be spuriously correlated with class labels [8].

This paper makes the following contributions:

- A complete end-to-end inspection pipeline integrating YOLO, a python-based data augmentation pipeline and a supervisory web dashboard.
- A controlled experimental comparison of augmented versus non-augmented training demonstrating the importance of domain-specific tuning and augmentation.
- A discussion of the system's broader applicability and its alignment with SCADA-like industrial control paradigms.

2. METHODOLOGY

2.1. System Architecture

The system is designed as a modular pipeline that integrates physical hardware, real-time inference software, and a supervisory web interface. Figure 1 illustrates the high-level architecture. The pipeline can be divided into four interconnected layers: the acquisition layer, the inference layer, the actuation layer, and the monitoring layer.

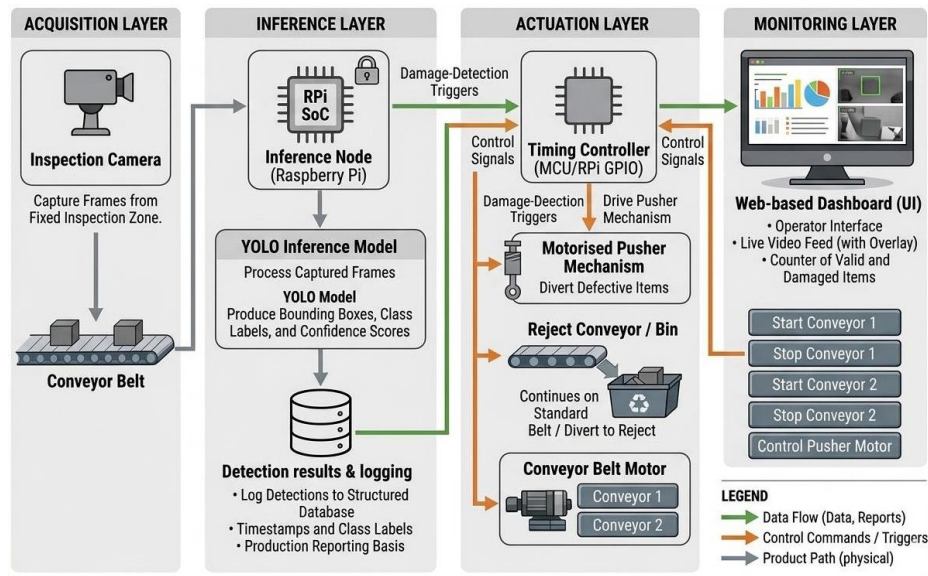


Figure 1: High-level system architecture diagram.

The acquisition layer captures frames from a camera positioned directly above the conveyor belt, framing a fixed inspection zone.

The inference layer passes each acquired frame to a YOLO model, producing bounding boxes, class labels, and confidence scores. Detection results are logged to a structured database with timestamps and class labels forming the basis for production reporting.

The actuation layer receives damage-detection triggers and drives a motorised pusher mechanism to divert defective items off the belt onto a reject conveyor or collection bin.

The monitoring layer provides a web-based dashboard displaying key performance indicators and serving as the operator interface. It provides a live video feed overlaid with detection bounding boxes, a real-time counter of valid and damaged items, and system control buttons.

2.2. Dataset and Augmentation

The custom dataset consists of images of boxes in two classes: structurally intact valid boxes and damaged (boxes exhibiting one or more of the following defects: crush deformation, edge tearing, or surface puncture).

Images are captured using a Pi camera operating in a fixed overhead position above a neutral background, mimicking the geometry of the production line inspection station. A minimum of 100 raw images per class are captured under different lighting conditions. The raw dataset is then annotated and expanded through the augmentation pipeline.

A Python script using the Albumentations library generates augmented training copies. All geometric transforms are applied jointly with annotations via Albumentations' BboxParams wrapper to maintain label validity. The augmentation pipeline was designed with two primary objectives: increasing effective dataset size and preventing the model from learning spurious correlations between damage and color.

The augmentation categories applied include geometric transformations such as random flips, rotations, cropping, and perspective distortion to simulate variations in product orientation. Color-space transformations involve hue shifts, saturation jitter, and brightness adjustments to prevent the model from associating specific colors with product classes. Noise and blur techniques inject Gaussian noise and simulate motion blur to replicate sensor conditions. Lastly, mosaic augmentation combines four images into one, aiding in detecting objects at smaller scales and in cluttered environments.

2.3. Model Training

YOLO is a single-stage anchor-free object detector that processes the full input image in a single forward pass. The nano variant is preferred for deployment on edge hardware with constrained compute. It provides the most practical combination of speed, accuracy, and deployment simplicity, making it ideal for real-time, resource-constrained applications like ours enabling detection of objects across a wide range of size.

Table 1 presents the key hyperparameters, their default values, the tuned values adopted for this project, and the rationale for each adjustment.

Table 1: YOLO hyperparameter configuration

Hyperparameter	Default Value	Tuned Value	Rationale
lr0 (initial LR)	0.01	0.005	Slower convergence for fine-grained features
hsv_h	0.015	0.05	Prevent color-damage co-learning
hsv_s	0.7	0.8	Increase saturation variance for colored boxes
hsv_v	0.4	0.5	Simulate varied factory lighting
conf (inference)	0.25	0.2	Reduce false positives on production line

The confidence threshold at inference time (conf) deserves particular attention in the context of industrial quality inspection. A false negative allows a defective product to reach the customer, which may have significant commercial or safety consequences. A false positive reduces throughput. The threshold should be selected based on the specific cost asymmetry of the deployment context. For most logistic applications, a slightly lower threshold (favoring recall over precision) is preferable.

2.4. Hardware Components

The system is built around three integrated hardware elements: a Raspberry Pi 5 paired with a camera module acting as the vision and processing unit, a motorized conveyor belt responsible for transporting boxes through the inspection zone, and an ejector actuator that physically diverts defective products upon a triggered signal. Together, these components form a closed-loop pipeline from intake to automated sorting as illustrated in figure 2.

Table 2: Hardware components and specifications

Component	Specification / Description	Role in System
Camera	Raspberry Pi Camera	Image acquisition for inference pipeline
Conveyor belt	Small-scale DC motor-driven belt	Transport products under inspection zone
Pusher mechanism	Rotary-to-linear conversion or pneumatic cylinder	Divert damaged items off primary belt
MCU / Controller	Raspberry Pi 5	Bridge inference output to physical actuation
Inference host		Run YOLO inference at real-time FPS

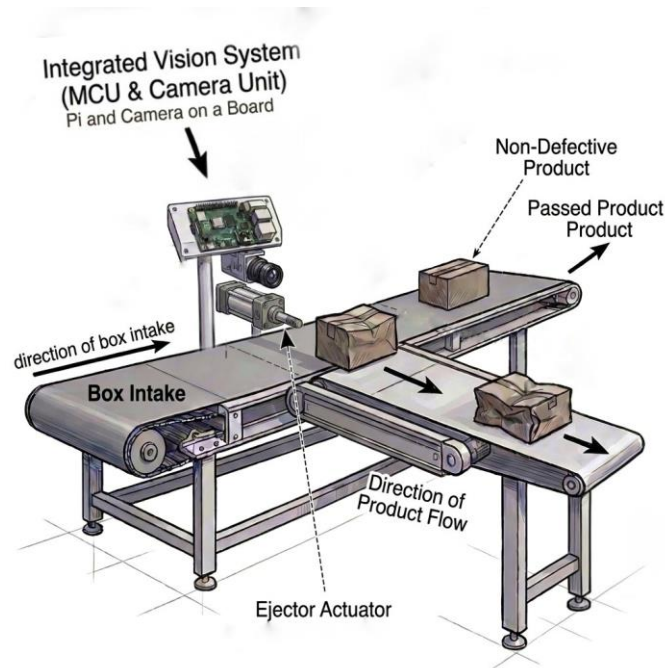


Figure 2: System overview of the automated defect detection and sorting setup.

2.5. Web Dashboard

The web dashboard is implemented as a full-stack web application. The backend serves the inference pipeline's output through a REST API. The frontend renders the live video feed with bounding box overlays, detection counters, and system controls using a responsive layout suitable for operation from both desktop workstations and mobile devices at the line.

This architecture aligns with modern industrial HMI design principles, employing high-performance graphic techniques like analog indicators over raw data. In an expanded deployment, the same interface could be extended to aggregate data from multiple production lines, monitor belt speed and actuator status, and provide alerts based on defect rate thresholds — functionality directly comparable to commercial SCADA systems.

An automated reporting module generates PDF documents summarizing inspection activities at daily, weekly, and monthly intervals. Each report includes the total number of inspected items, counts and percentages of damaged items, hourly detection volume charts, and system uptime statistics. These reports, created using the Python PDF library ReportLab, can be downloaded from the dashboard. The automated generation provides a continuous quality audit trail and supports data-driven process improvements by enabling management to track long-term defect rate trends without manual data extraction.

3. RESULTS

3.1. Experimental Setup

The dataset was split into training, validation, and test subsets in an 80:10:10 ratio. Training was conducted using the YOLO11n architecture due to its suitability for resource-constrained deployment. The augmented configuration applied standard techniques including flipping, scaling, color jitter, random brightness contrast, motion blur and gaussian noise, while the baseline model was trained without augmentation. All experiments were run for 100 epochs under identical training conditions to ensure a fair comparison.

3.2. Training Convergence

The training and validation losses were monitored over 100 epochs to verify stable convergence and the absence of overfitting in both configurations.

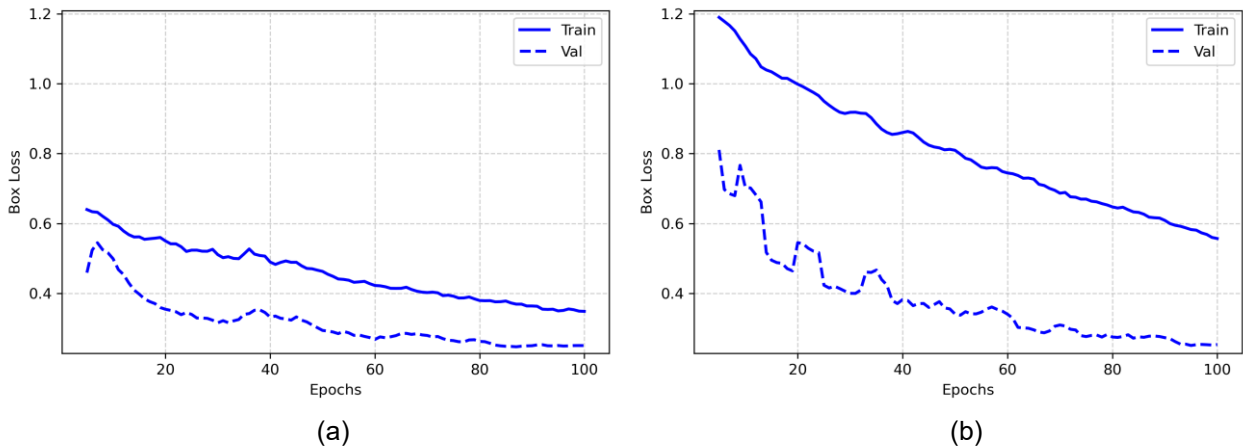


Figure 3: Training and validation box loss over 100 epochs for (a) baseline and (b) augmented models.

Figure 3 presents the training and validation box loss for the baseline (a) and augmented (b) models. The baseline model begins from a moderate initial loss and descends smoothly throughout training, with the validation loss closely tracking the training loss and both converging to a stable minimum by epoch 100. The augmented model begins from a noticeably higher initial box loss, which is expected due to the geometric transformations applied during augmentation, including random rotations, perspective distortion, and scale variation, presenting the model with a substantially harder localization task from the first epoch, as bounding box regression must account for a wider range of object positions and shapes. Despite this harder start, the augmented model converges steadily without divergence, and the absence of a widening gap between its training and validation curves confirms that it generalises well to the held-out validation set with no signs of overfitting.

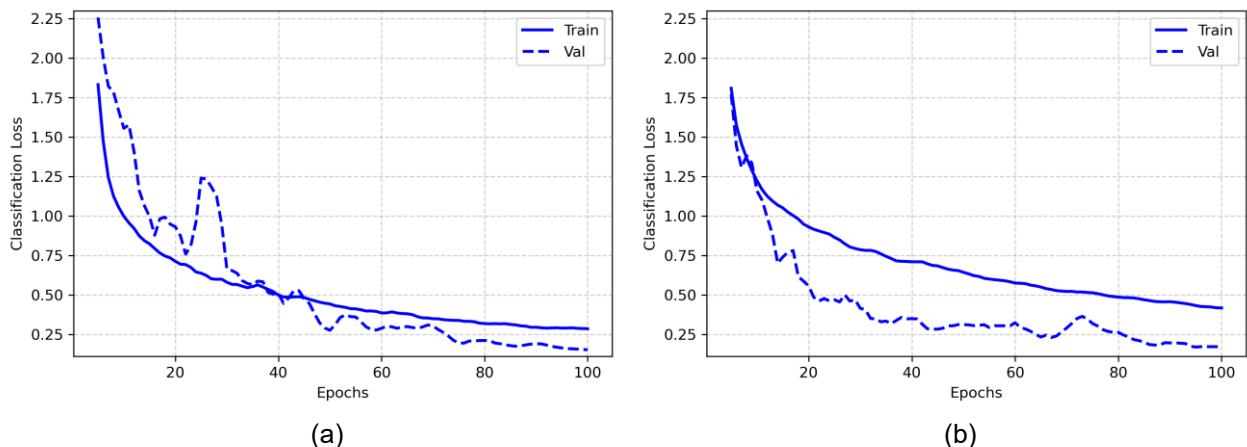


Figure 4: Training and validation class loss over 100 epochs for (a) baseline and (b) augmented models.

Figure 4 presents the training and validation classification loss for the baseline (a) and augmented (b) models. The baseline model begins with a sharp initial drop, but its validation loss exhibits a pronounced spike around epoch 20 before recovering. This instability reflects the model encountering the limited visual diversity of the non-augmented training set, having rapidly fitted to a narrow distribution, it momentarily struggles to generalise when evaluated on the validation set. The augmented model also begins from a high initial classification loss and shows some early oscillation in the validation curve, but the fluctuations are notably less severe, and the validation loss tracks the training loss more consistently throughout the remainder of training.

The comparative stability is a direct consequence of the augmented training distribution, which exposes the model to color-shifted, geometrically varied, and noise-corrupted samples from the outset, preventing premature convergence to a narrow set of visual patterns. While both models reach similar final classification loss values, the smoother optimisation trajectory of the augmented model represents a meaningful reliability advantage for production deployment, where training stability is an indicator of robustness to unseen variation.

3.3. Detection Performance

The mAP progression over training epochs is shown for both models on shared axes to facilitate direct comparison.

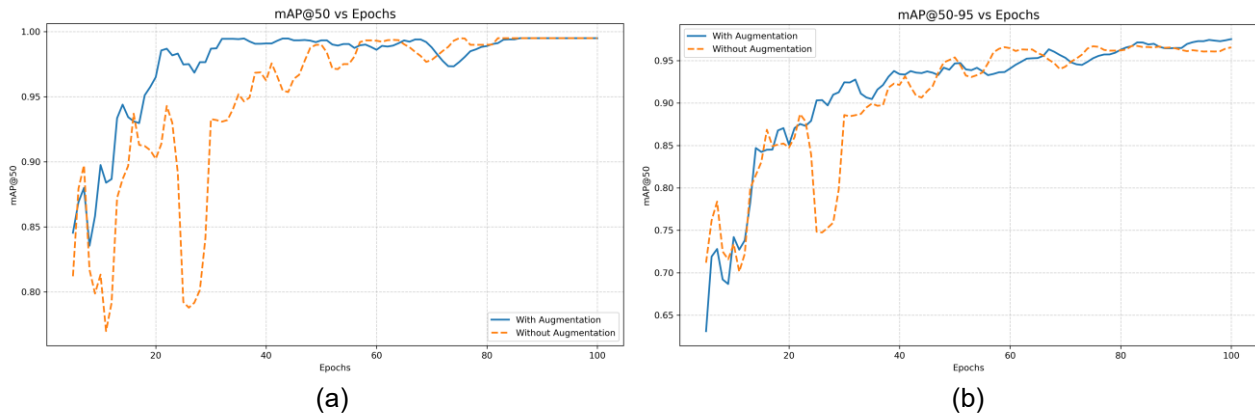


Figure 5a: mAP@50 over 100 training epochs for both models overlaid.

Figure 5b: mAP@50-95 over 100 training epochs for both models overlaid.

Both models reach an mAP@50 of 0.995, confirming that augmentation does not compromise final classification accuracy. The augmented model, however, reaches this plateau later, consistent with the slower but more regularized convergence imposed by the augmented training distribution.

The augmented model achieves a final mAP@50-95 of 0.9765 compared to 0.9667 for the baseline. This metric penalizes imprecise bounding box localization more heavily than mAP@50, and the gap between the two curves reflects the improvement in box localization quality attributable to the geometric augmentations, which expose the model to a wider range of object scales, orientations, and positions during training.

3.4. Quantitative Comparison

Table 3: Final validation metrics at epoch 100 for both training configurations

Metric	With Augmentation	Without Augmentation
Precision	0.9965	0.9961
Recall	1.0000	1.0000
mAP@50	0.9950	0.9950
mAP@50-95	0.9765	0.9667
Val Box Loss	0.2483	0.2521
Val Cls Loss	0.1690	0.1512
Training Time	2577.7	753.6

Table 3 summarises the final validation metrics for both models at epoch 100. The two models reach identical mAP@50 and recall; the augmented model leads on mAP@50-95 by 0.0098, indicating superior bounding box localisation. The most practically significant difference is training time: the baseline converges in 753.6 seconds compared to 2577.7 seconds for the augmented model, reflecting the increased effective dataset size introduced by augmentation.

3.5. Confusion Matrices

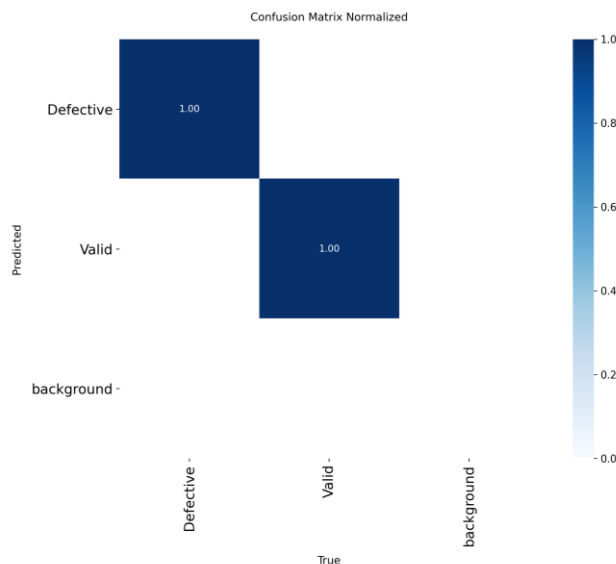


Figure 6: Normalised confusion matrix for the augmented model.

The augmented model correctly classifies every instance in the test set across both classes, with no off-diagonal entries. This perfect classification behavior at the chosen confidence threshold is consistent with the recall of 1.0 and precision of 0.9965 reported in Table 1.

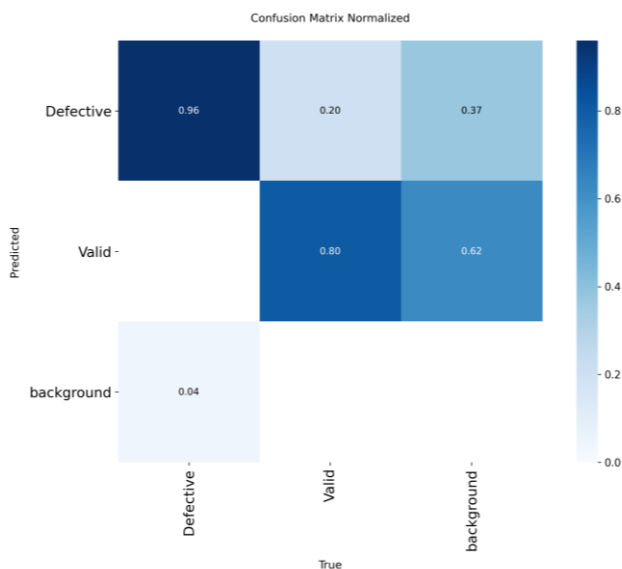


Figure 7: Normalized confusion matrix for the baseline model.

The baseline matrix reveals minor misclassifications absent from the augmented model's matrix, despite the two models sharing the same recall and mAP@50 values. This distinction — invisible in the aggregated validation metrics — is the clearest visual evidence that augmentation produces a more reliable classifier.

4. DISCUSSION

The high performance is expected within the controlled deployment context of this system and reflects a deliberate design choice rather than an overfit result. Unlike general-purpose object detection models that must generalize across thousands of categories and unconstrained environments, an industrial inspection model is trained and deployed to solve one specific problem e.g., classifying boxes as valid or damaged on a defined conveyor line. High performance on that narrow domain is the objective, not a concern. Generalization beyond the target domain is neither required nor desirable as introducing that trade-off would reduce accuracy on the only task the system is asked to perform.

Crucially, the test set itself already encompasses the meaningful sources of variation present in the deployment environment. Images were captured under distinct lighting conditions and include boxes at varied orientations and positions within the camera frame. The model therefore does generalize, but appropriately across the geometric and illumination variation it will encounter in production, not across unrelated product categories it will never see.

The confusion matrices (Figures 6 and 7) reveal a distinction that the aggregated validation metrics in Table 3 obscure. Both models achieve identical recall and mAP@50, which could suggest augmentation provides no benefit at the classification level. The matrices show otherwise: the augmented model produces zero misclassifications across the entire test set, while the baseline model records a small number of incorrect predictions. This difference is the most direct visual evidence that augmentation produces a more reliable classifier, and it demonstrates the value of examining confusion matrices alongside summary statistics rather than relying on aggregate metrics alone.

The augmented model's superior mAP@50-95 score of 0.9765 compared to 0.9667 for the baseline reflects tighter bounding box localization, attributable to the geometric transformations in the augmentation pipeline. In the current system this distinction is operationally relevant if bounding box coordinates are used to characterize defect extent or position.

The implication of the augmentation ablation study generalizes beyond this case study, any industrial inspection system in which product color varies across the dataset, whether due to packaging variants, seasonal labelling, or incidental markings, should apply aggressive HSV augmentation and verify that recall is uniform across color subsets of the test set.

The web dashboard, combined with the implemented automated PDF reporting module, completes the system's role as a SCADA analogue. The reporting module generates daily, weekly, and monthly PDF documents covering total inspection volume, defect counts and rates, hourly detection charts, and system uptime. These reports provide a persistent quality audit trail and make long-term defect rate trends visible to management without manual data extraction — a capability that closes the gap between the vision system and the plant-level quality management infrastructure it is designed to integrate with.

5. CONCLUSION AND FUTURE WORK

This paper presented a complete computer vision-based quality inspection system for industrial conveyor applications, using YOLO as the detection backbone and paper box classification as the case study. The system integrates real-time inference, and a web based supervisory dashboard into a cohesive pipeline demonstrating that a single camera and a trained neural network can replicate much of the functionality provided by arrays of dedicated sensors at significantly lower cost and with greater adaptability.

The augmentation study confirmed that color-space augmentation is particularly important for datasets with high color variance, while the cross-domain study illustrated the need for targeted hyperparameter re-tuning when deploying the pipeline to new product domains. The web dashboard, designed with SCADA principles in mind, provides operators with real-time monitoring, remote control, and automated production reporting suitable for integration with plant-level quality management systems.

The results demonstrate that computer vision-based inspection is not merely a research curiosity but a practically deployable technology for industrial quality control, and that the system described in this paper provides a replicable and extensible foundation for future work.

Future work will focus on completing end-to-end hardware validation at multiple belt speeds, evaluating system robustness under variable industrial lighting and higher conveyor speeds, and exploring integration with plant-level protocols such as OPC-UA and MQTT.

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